INCH-POUND
MIL-PRF-3098/109F
w/AMENDMENT 1
7 October 2019
SUPERSEDING
MIL-PRF-3098/109F
7 January 2010

PERFORMANCE SPECIFICATION SHEET

CRYSTAL UNIT, QUARTZ, CR130/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-3098.

Pertinent characteristics: 0.8 MHz to 20 MHz; fundamental; noncontrolled; antiresonance.

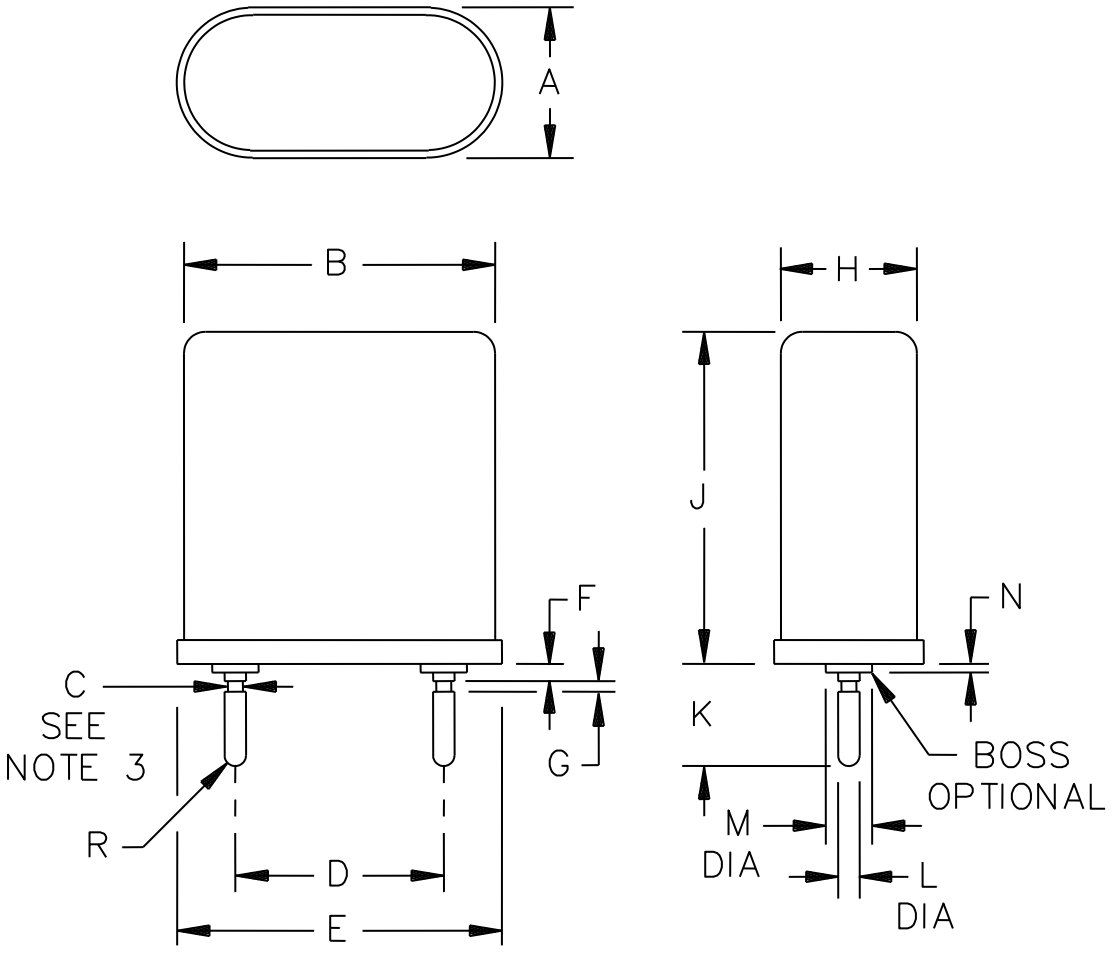


FIGURE 1. Crystal unit – CR130/U.

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Ltr	Inches		mm	
	Min	Max	Min	Max
A	---	.352	---	8.94
B	---	.725	---	18.41
C	.030	.037	0.76	0.94
D	.460	.476	11.68	12.09
E	---	.757	---	19.23
F	.010	.070	0.25	1.78
G	.015	.025	0.38	0.63
H	---	.317	---	8.05
J	---	.775	---	19.68
K	.223	.248	5.66	6.30
L	.066	.216	1.68	5.49
M	.048	.052	1.22	1.32
N	.015	.025	0.38	0.63

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. The pin undercut may be omitted.
4. Marking to be in accordance with MIL-PRF-3098.

FIGURE 1. Crystal unit – CR130/U - Continued.

REQUIREMENTS:

Dimensions, marking, and configuration: See figure 1.

Frequency range: 0.8 MHz to 20.0 MHz, inclusive.

Frequency tolerance (operating temperature range):

Primary: ±20 parts per million (ppm).

Secondary: ±30 ppm.

Equivalent resistance: See table I.

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TABLE I. Equivalent resistance.

Frequency range, inclusive	Maximum resistance
MHz	Ohms
0.80 to 0.85	620
0.85+ to 0.90	600
0.90+ to 1.00	570
1.00+ to 1.12	540
1.12+ to 1.25	490
1.25+ to 1.37	450
1.37+ to 1.50	410
1.50+ to 1.62	370
1.62+ to 1.75	330
1.75+ to 1.87	300
1.87+ to 2.00	290
2.00+ to 2.12	270
2.12+ to 2.25	240
2.25+ to 2.60	190
2.60+ to 3.00	150
3.00+ to 3.40	110
3.40+ to 3.75	90
3.75+ to 4.00	75
4.00+ to 5.00	60
5.00+ to 7.00	35
7.00+ to 10.00	24
10.00+ to 15.00	22
15.00+ to 20.00	20

Mode of oscillation: Fundamental.

Temperature ranges:

Primary: -40°C to +90°C, inclusive.

Secondary: -55°C to -40°C and +90°C to +105°C, inclusive.

Rated drive level: 1.0 mW maximum.

Capacitance, shunt: 7.0 pF, maximum.

Antiresonance: load capacitance: 30.0 pF ±0.5 pF. 1/

1/ This crystal unit is similar to Crystal Type CR-85/U, except for operation at antiresonance and maximum equivalent resistance.

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Shock (specified pulse):	0.8 to 2.0 MHz	2.0+ to 20.0 MHz
Frequency change permitted:	±10 ppm	±5 ppm
Equivalent resistance change permitted:	±15 percent	±10 percent

Vibration: MIL-STD-202-204, test condition A.	0.8 to 2.0 MHz	2.0+ to 20.0 MHz
Frequency change permitted:	±10 ppm	±5 ppm
Equivalent resistance change permitted:	±15 percent	±10 percent

Thermal shock:	0.8 to 2.0 MHz	2.0+ to 20.0 MHz
Frequency change permitted:	±10 ppm	±5 ppm
Equivalent resistance change permitted:	±15 percent	±10 percent

Aging:

Frequency change permitted: ±5 ppm

Part or Identifying Number (PIN): CR130/U (followed by specified frequency), see 1.2 of MIL-PRF-3098.

Reference documents. In addition to MIL-PRF-3098, this document references the following:

MIL-STD-202-204

Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

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Custodians:
Army - CR
Navy - EC
Air Force - 85
DLA - CC

Review activities:
Army - AR, MI, SM
Navy - AS, MC, SH
Air Force - 19

Preparing activity:
Army - CR

Agent:
DLA - CC

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